

Spin, Vorticity, and N-Body Dynamics in a Superfluid Defect Toy Model

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Abstract

We extend our superfluid-defect toy universe to spin and full N -body dynamics at 1PN order. The vacuum is a compressible superfluid and massive bodies are flux-tube “throat” defects that exchange mass/flux with the medium (effective sinks in the 3D description). Earlier papers fixed the orbital sector ($\beta = 3$) [1] and selected a stiff $n = 5$ equation of state from optical 1PN observables [2].

Here we promote defects to composite “dyons” (sink + vortex ring). Their far-field circulation defines a gravitomagnetic vector potential with the correct J/r^3 scaling, reproducing the Lense-Thirring effect and fixing the vortex-spin calibration by matching the Kerr weak-field limit. In the N -body problem, density-dependent masses reproduce the static G^2 three-body term of the Einstein-Infeld-Hoffmann (EIH) Lagrangian. The remaining velocity-dependent EIH cross terms arise from overlap energy of translational wakes. Using an isotropic projector decomposition into longitudinal and transverse solenoidal (vortical) components (with an optional helical mode), we match the full EIH cross-term tensor with real parameters: $a_H = 0$, $\alpha^2 = 3/4$, and $K = 2/\pi^2$. The positive value $\alpha^2 = 3/4$ makes the quadratic wake functional positive-definite, showing that earlier imaginary couplings were artifacts of an incomplete wake basis. Subject to the dynamical requirement that moving sink defects generate a long-range wake $u \propto 1/r^2$ (equivalently $u(\mathbf{k}) \propto 1/k$), the model reproduces GR’s 1PN scalar, optical, spin, and vector dynamics while remaining consistent with the calibrated orbital sector and optical tests.

1 Introduction

1.1 Motivation and overview

The classic solar-system tests of gravity—perihelion precession, light bending, Shapiro time delay, gravitational redshift, and the dynamics of weakly bound N -body systems—are often summarized in a single statement: the Schwarzschild solution of General Relativity (GR) with post-Newtonian (PN) parameters $\beta_{\text{PPN}} = \gamma_{\text{PPN}} = 1$ passes all currently accessible 1PN tests. Throughout this series we use β_{PPN} and γ_{PPN} as shorthand for the values inferred from the standard 1PN observables. For optical tests we denote the optically inferred value by γ_{obs} and identify $\gamma_{\text{PPN}} \equiv \gamma_{\text{obs}}$. In the toy model this observational equivalence does not require a single underlying bare metric: light and matter couple to distinct acoustic and hydrodynamic projections that are calibrated to agree at 1PN. From this vantage point, any alternative description of gravity must either reproduce the Schwarzschild metric in the appropriate weak-field limit or offer a comparably constrained and falsifiable mechanism by which the same observables arise.

This paper continues a different line of attack. Instead of starting from a Lorentzian spacetime and quantizing perturbations of the metric, we treat gravity as an *emergent* phenomenon in a “toy universe” where the vacuum is a compressible superfluid and massive bodies are flux-tube defects that drain this vacuum. The effective gravitational dynamics experienced by defects are then encoded in the density, pressure, and flow of the surrounding superfluid, together with a small number of phenomenological parameters that characterize the throat geometry and equation of state. In this language, the usual PN “potential” and its higher-order corrections are realized as different facets of a single hydrodynamic configuration.

Most analogue-gravity constructions are qualitative: they reproduce some aspect of GR kinematics (for example, horizon structure or redshift) without attempting a quantitative match to the full 1PN phenomenology of the solar system. In this series of papers we pursue a more aggressive question in a deliberately simple setting:

How far toward the full 1PN phenomenology of GR can one push a minimal, classical hydrodynamic toy model of defects in a compressible superfluid?

Paper I and Paper II showed that, with a suitable choice of scalar lag dynamics and equation of state, the toy model can already reproduce the standard scalar and optical 1PN tests. The present work addresses the remaining ingredients: spin-induced gravitomagnetism (Lense–Thirring) and the full N -body interaction encoded in the Einstein–Infeld–Hoffmann (EIH) Lagrangian. Our goal is not only to match the GR coefficients but to understand which features of the emergent superfluid description are *forced* by that match.

1.2 Summary of Papers I and II

Paper I in this series [1] developed the *orbital* sector of the toy universe. The vacuum was modeled as a compressible superfluid, and massive bodies as flux-tube “throats” of radius a and depth L that drain the surrounding fluid. A scalar “lag” field allowed the bulk fluid to slip relative to the defects, and a position-dependent kinetic prefactor $\sigma(r)$ encoded how defect inertia is renormalized in the throat background. The long-range field sourced by a defect naturally split into two scalar contributions: an instantaneous Poisson sector that reproduces the Newtonian $1/r$ potential, and a retarded lag sector that supplies the finite-propagation effects. In the updated calibration, the scalar lag field relaxes in the static single-body limit, so the 1PN perihelion precession is generated entirely by the inertia profile $\sigma(r)$. Demanding that nearly Keplerian orbits around an isolated defect reproduce the GR 1PN perihelion advance fixes a single dimensionless parameter $\beta = 3$ in the defect Lagrangian. The same cavitation analysis that ties the defect mass to the surrounding density implies a density-depletion coefficient $\kappa_\rho = 1$, which controls the static G^2 “gravity gravitates” term in the N -body problem.

Paper II [2] extended the toy universe to the *optical* sector. Treating the vacuum as a stiff ($n = 5$) polytropic superfluid endowed the medium with a density-dependent sound speed and hence a radial refractive index profile $N(r)$ around a defect. By constructing an effective optical metric from $N(r)$ and comparing to the Schwarzschild metric, the analysis showed that $n = 5$ is uniquely selected (within the class considered) by the requirements of matching 1PN light bending, Shapiro time delay, and gravitational redshift. As emphasized in Paper II, this agreement is enforced at the level of calibrated 1PN observables (light bending, Shapiro delay, redshift, and perihelion precession), while the underlying acoustic (light) and hydrodynamic (matter) projections can differ (bi-metric structure). Once the scalar and optical contributions are combined and calibrated, the inferred PPN parameters satisfy $\beta_{\text{PPN}} = \gamma_{\text{PPN}} = 1$, so the toy model reproduces both the orbital

and optical 1PN tests with no new free parameters beyond those already fixed by the scalar and optical sectors.

Taken together, Papers I and II establish that a single superfluid-defect toy model can account for the scalar and optical 1PN tests of gravity with a small set of tightly constrained parameters: the throat aspect ratio L/a , the scalar renormalization parameter β , and the polytropic index n . What remains, and is addressed in the present work, are the *vector* phenomena associated with spin and N -body dynamics.

1.3 Scope and roadmap

At the 1PN level, the missing pieces fall into two closely related categories. First, a spinning gravitating body generates a *gravitomagnetic* field: in GR this is encoded in the off-diagonal metric components g_{0i} and observed as the Lense–Thirring precession of gyroscopes and orbital planes around rotating masses. Second, the dynamics of multiple bodies at 1PN order are governed by the Einstein–Infeld–Hoffmann Lagrangian, which contains not only the Newtonian pairwise potential but also static G^2 three-body terms and velocity-dependent interaction terms with a very specific tensor structure. Reproducing these ingredients is a stringent test of any emergent-gravity model.

In the superfluid language, both effects are naturally associated with *flow*. We model spinning defects as *dyons*: composite objects in which a flux-tube sink (mass) is bound to a localized vortical core (spin) in the surrounding superfluid. The far-zone flow is taken to have a dipole-like (vector-potential) structure $\mathbf{v} \propto (\mathbf{J} \times \hat{\mathbf{r}})/r^2$ (so Cartesian components scale as $1/r^2$); through the metric dictionary it reproduces the weak-field Kerr gravitomagnetic potential and the observable Lense–Thirring precession $\Omega_{\text{LT}} \propto 1/r^3$. At the same time, the overlapping velocity fields of multiple dyons give rise to an effective vector interaction that scales as $1/r$ and can, in principle, be matched to the EIH velocity-dependent terms.

Section 2 reviews the ingredients we import from Papers I and II: the scalar lag field, the position-dependent inertia, the $n = 5$ stiff equation of state, and the dictionary that maps density and flow to metric components. Section 3 introduces the dyon construction and shows that it reproduces the GR Lense–Thirring effect with a fixed calibration between the vortex strength and the physical angular momentum J . Section 4 then turns to the N -body problem: we show how the density-dependent mass generates the static G^2 three-body term, construct the vector interaction from overlapping translational wakes, and derive the conditions under which the resulting tensor structure matches the EIH Lagrangian. In parallel with the scalar ($q = 1$) [1] and optical ($n = 5$) [2] sectors fixed in Papers I and II, the EIH cross-term coefficients alone fix the translational wake that supplies the mixed-velocity terms: the isotropic projector decomposition yields a real wake mixing $\alpha^2 = 3/4$ (with minimal $a_H = 0$) and overall coupling $K = 2/\pi^2$. Rather than signaling an instability, the multi-body sector thus selects a specific hydrodynamic response for the vacuum.

Finally, Section 5 summarizes how Papers I and II together with the present work reproduce the full 1PN phenomenology of GR within the toy superfluid universe, discusses the limitations of this construction, and outlines directions for extending the model to higher PN orders, radiative effects, and the electromagnetic sector.

2 Inputs from Papers I and II: Scalar Sector and Metric Dictionary

In this section we collect the minimal ingredients from the orbital and optical analyses of Papers I and II that will be treated as inputs for the present work. Our goal is not to re-derive those results,

but to make explicit which structures are assumed, which parameters have already been fixed, and how they combine into an effective metric dictionary that will be extended to include spin and N -body dynamics in the sections that follow.

2.1 Superfluid defect toy model recap

The underlying ontology of the toy universe is unchanged from Papers I and II. The fundamental medium is a homogeneous, compressible superfluid with bulk mass density ρ_0 and characteristic wave speed c_s . Matter is represented by localized *defects* that act as sinks of the superfluid: each defect removes fluid from the bulk and routes it along a narrow “throat” of radius a and depth L , before returning it to the ambient medium. On scales large compared to a and L the details of the throat geometry are coarse-grained into an effective point-like source of strength

$$\mu \equiv GM, \quad (1)$$

where M is the inertial mass associated with the defect and G is the effective gravitational constant in the toy model.

The superfluid bulk is described by a density field $\rho(\mathbf{x}, t)$, a pressure $p(\mathbf{x}, t)$, and a velocity field $\mathbf{v}(\mathbf{x}, t)$ which obey the usual continuity and Euler equations, augmented by sink terms localized on the defect cores. In the simplest, non-rotating sector considered in Paper I the flow is irrotational and can be written in terms of a scalar potential $\Phi(\mathbf{x}, t)$ whose gradient gives the acceleration of a test defect,

$$\mathbf{a}(\mathbf{x}, t) = -\nabla\Phi(\mathbf{x}, t). \quad (2)$$

For a static defect the far-field solution reduces to the familiar Newtonian form $\Phi(r) \simeq -\mu/r$. More generally, time dependence and finite propagation speed in the medium give rise to a retarded scalar contribution which behaves as a post-Newtonian correction to the effective potential.

In the full toy universe a complete defect (a “dyon”) can in principle carry both a scalar sink and a vector vortex; in Papers I and II the focus was on the scalar and optical sectors, and the vector (spin) structure was left implicit. In the present work we will restore the dyon picture explicitly, but the scalar fields and equation of state remain exactly those determined in the earlier papers.

2.2 Scalar lag field, inertia profile, and β

The scalar sector of the toy model contains two distinct pieces. The first is an instantaneous Poisson contribution $\Phi_N(r) = -\mu/r$ sourced by the defect, which plays the role of the Newtonian potential. The second is a retarded “lag” contribution Φ_{lag} arising from finite propagation speed in the medium, which obeys a wave equation of the schematic form

$$\frac{\partial^2 \Phi}{\partial t^2}(\mathbf{x}, t) = c_s^2 [\nabla^2 \Phi(\mathbf{x}, t) - 4\pi G \rho(\mathbf{x}, t)]. \quad (3)$$

For slowly moving sources this retarded contribution reduces, at leading post-Newtonian order, to a spherically symmetric $1/r^2$ correction,

$$\Phi_{\text{lag}}(r) = -\frac{\mu^2}{2c_s^2 r^2}, \quad (4)$$

so that the total effective potential seen by a non-relativistic test defect is

$$\Phi_{\text{eff}}(r) = \Phi_N(r) + \Phi_{\text{lag}}(r) = -\frac{\mu}{r} - \frac{\mu^2}{2c_s^2 r^2}. \quad (5)$$

Paper I further introduced a position-dependent kinetic prefactor $\sigma(r)$ which encodes how the inertial response of a defect is renormalized by the throat background. At the level of an effective point-particle description, the kinetic term acquires a multiplicative factor $[1 + \sigma(r)]$, and the inertial mass m appearing in the Lagrangian is replaced by

$$m_{\text{eff}}(r) = m [1 + \sigma(r)], \quad (6)$$

with

$$\sigma(r) = \beta \frac{\mu}{c_s^2 r}, \quad (7)$$

where β is a dimensionless constant. One can think of $\sigma(r)$ as a simple model for a radial dependence of the spatial metric components experienced by the defect, or equivalently as a phenomenological way of encoding hydrodynamic inertia in the coarse-grained description.

When the Lagrangian with $\Phi_{\text{eff}}(r)$ and $\sigma(r)$ is expanded consistently to 1PN order and the resulting equations of motion are applied to nearly Keplerian orbits, the scalar lag field relaxes in the static single-body limit and does not contribute to the perihelion advance. The entire 1PN precession is generated by the position-dependent inertial dressing encoded in $\sigma(r)$, yielding a total coefficient

$$\Delta\varphi_{\text{tot}} = 6 \frac{\pi\mu}{c_s^2 a(1 - e^2)}, \quad (8)$$

with a and e the orbital semi-major axis and eccentricity, provided

$$\beta = 3. \quad (9)$$

In this sense, Paper I fixed the single scalar parameter β by requiring that the toy model reproduce the GR 1PN perihelion precession purely through inertial renormalization. This same value of β will be assumed throughout the present work, and will enter the N -body analysis via the density-dependent mass and static G^2 term in the EIH Lagrangian.

2.3 Optical sector and the stiff $n = 5$ vacuum

Paper II turned to the optical and clock sectors of the toy model. There the superfluid vacuum was endowed with a polytropic equation of state

$$p = K \rho^{1+1/n}, \quad (10)$$

with n the polytropic index and K a constant. A mass defect was again modeled as a flux-tube sink embedded in this vacuum, with the same coarse-grained parameter $\mu = GM$ describing its far-field strength. Hydrostatic balance in the presence of the effective potential $\Phi(r)$ then implies a radial density and pressure deficit around the defect, which in turn modify the local sound speed $c_s(r)$ and define a refractive index profile

$$N(r) = \frac{c_0}{c_s(r)}, \quad (11)$$

where c_0 is the sound speed in the far-field vacuum.

For a general polytropic index n one obtains, in the weak-field regime, a refractive index of the form

$$N(r) \simeq 1 + \alpha_n \frac{GM}{c_0^2 r}, \quad (12)$$

with a coefficient α_n that depends on n . Paper II showed that the stiff $n = 5$ branch is singled out by the 1PN optical tests. Specializing to $n = 5$ yields

$$N_{n=5}(r) \simeq 1 + 2 \frac{GM}{c_0^2 r}, \quad (13)$$

which is the profile used in the lensing and Shapiro-delay calculations. In the 1PN matching limit one identifies c_0 with c , so the coefficient in Eq. (13) is directly comparable to the GR result.

Treating lightlike excitations as rays propagating through an inhomogeneous medium with refractive index $N(r)$, Paper II computed the weak-field bending angle for a point mass and showed that Eq. (13) reproduces the standard GR deflection angle with the correct coefficient. An analogous analysis of signal propagation time in the same index profile yielded the familiar logarithmic Shapiro delay, again with the GR coefficient. Finally, by constructing an effective optical metric from $N(r)$ and relating it to the PN expansion of the spacetime metric, Paper II showed that the combined scalar and optical sectors correspond to PPN parameters

$$\beta_{\text{PPN}} = 1, \quad \gamma_{\text{PPN}} = 1, \quad (14)$$

when all contributions are accounted for. The toy model therefore reproduces both the orbital and optical 1PN tests of GR with a single choice of throat geometry and equation of state: a flux-tube defect with a fixed aspect ratio L/a , scalar renormalization parameter $\beta = 3$ in the defect Lagrangian, and a stiff $n = 5$ polytropic vacuum.

2.4 Remaining 1PN tasks

The inputs summarized above completely determine the scalar and optical sectors of the toy universe at 1PN order. The Newtonian potential and its scalar lag correction fix the effective g_{00} components relevant for slow-motion dynamics, while the refractive index profile in the $n = 5$ vacuum captures the optical manifestations of the spatial metric and yields $\gamma = 1$ when compared to GR. From the PPN point of view, the model already behaves like GR in the monopole, spherically symmetric sector.

What remains are the genuinely *vector* phenomena associated with flow and vorticity. A spinning mass generates a gravitomagnetic field, encoded in GR by the off-diagonal metric components g_{0i} and observed as Lense–Thirring precession. In an N -body system, the full 1PN dynamics are captured by the Einstein–Infeld–Hoffmann Lagrangian, which contains static G^2 three-body terms and velocity-dependent pairwise terms with a highly constrained tensor structure. In the superfluid language these effects arise from the velocity field $\mathbf{v}(\mathbf{x}, t)$ sourced by spinning defects and from the interaction energy of overlapping flows.

The present paper builds on the scalar potential Φ_{eff} , the inertia profile $\sigma(r)$, and the $n = 5$ refractive index $N(r)$ summarized above, and extends the toy model to include spin and N -body dynamics. In particular, we will construct composite defects (dyons) whose scalar sink and vortex ring together reproduce the GR Lense–Thirring field, and we will show how the interaction of their flows generates the full EIH 1PN Lagrangian, fixing the translational wake parameters to ($a_H = 0$, $\alpha^2 = 3/4$, $K = 2/\pi^2$) once the $q = 1$ scalar background from Paper I [1] and stiff $n = 5$ optical sector from Paper II [2] are taken into account.

3 Spin and the Lense–Thirring Effect

In Papers I and II the defects were treated as non-spinning sources: only the scalar sink structure of the throat entered the analysis. From the GR point of view this corresponds to working with a

Schwarzschild-like sector where the metric is diagonal and the off-diagonal components g_{0i} vanish. The next natural step is to endow the defects with spin and ask whether the same superfluid toy universe can reproduce the gravitomagnetic phenomena associated with rotating masses, most notably the Lense–Thirring effect. In this section we show that a composite “dyon” defect—a flux-tube sink bound to a vortex ring in the surrounding superfluid—does exactly that.

3.1 The radial mismatch problem

In GR, the dominant spin effect of an isolated, slowly rotating body with angular momentum vector $\mathbf{J}$ is described by the Lense–Thirring precession. To leading order in J and in the weak-field limit, the metric can be written as

$$g_{0i}^{(\text{GR})} = -\frac{2G}{c^3} \epsilon_{ijk} \frac{J^j x^k}{r^3} + \mathcal{O}(J^2), \quad (15)$$

with $r = |\mathbf{x}|$. The precession of a gyroscope at position $\mathbf{r}$ is then governed by the gravitomagnetic precession vector

$$\boldsymbol{\Omega}_{\text{LT}}(\mathbf{r}) = \frac{G}{c^2 r^3} [3(\mathbf{J} \cdot \hat{\mathbf{r}}) \hat{\mathbf{r}} - \mathbf{J}], \quad (16)$$

which scales as J/r^3 . Any viable emergent-gravity model must reproduce this $1/r^3$ behavior together with the angular dependence encoded in Eq. (16).

A natural first guess in a superfluid picture is to model a spinning defect as a line vortex aligned with $\mathbf{J}$. For a straight vortex along the z -axis the azimuthal velocity in cylindrical coordinates $(r_\perp, \phi, z)$ is

$$v_\phi(r_\perp) = \frac{\Gamma}{2\pi r_\perp}, \quad (17)$$

where Γ is the circulation. The local angular velocity of the fluid around the vortex is then $\omega \sim v_\phi/r_\perp \propto 1/r_\perp^2$. This has the wrong radial dependence: the induced precession falls as $1/r^2$ rather than $1/r^3$ and cannot be made compatible with the Lense–Thirring scaling at large distances. Moreover, the flow is topologically constrained to be purely azimuthal; it does not reproduce the dipolar angular structure in Eq. (16).

A different classical construction is to consider a slowly rotating sphere of radius R and angular velocity $\boldsymbol{\Omega}$ embedded in an otherwise static fluid. Continuity then demands a compensating backflow around the sphere, and in the far field the velocity field takes the form of a dipole:

$$\mathbf{v}_{\text{backflow}}(\mathbf{r}) \sim \frac{R^3}{r^3} \boldsymbol{\Omega} \times \mathbf{r}, \quad r \gg R. \quad (18)$$

This has the desired dipolar structure $\propto \boldsymbol{\Omega} \times \mathbf{r}/r^3$ (Cartesian components $\propto 1/r^2$) and the right $\boldsymbol{\Omega} \times \mathbf{r}$ angular dependence to mimic the GR gravitomagnetic vector potential locally. However, when used as the building block for an N -body interaction it behaves too much like a rigid rotation of the bulk: the overlap energy of two such backflows decays as $1/r^3$ and therefore induces an interaction that falls too rapidly with separation to match the Einstein–Infeld–Hoffmann (EIH) vector term, which requires an effective $1/r$ potential. We will return to this tension in Sec. 4; for now it suffices to note that neither a simple line vortex nor the rigid backflow of a spinning sphere provides a satisfactory starting point for a unified description of spin and N -body dynamics.

3.2 The dyon solution

The construction that succeeds in the toy universe is a composite defect, or *dyon*, in which a scalar sink and a vortex ring are bound together on the same throat. The scalar sink is exactly the one used in Papers I and II: it removes fluid from the bulk and sources the effective potential $\Phi(r)$. The new ingredient is a localized vortical core that carries the defect's spin and sources a long-range azimuthal flow. In this paper we take the far-zone behavior of that flow as a phenomenological requirement imposed by matching the weak-field Kerr gravitomagnetic potential (equivalently the Lense–Thirring observable); the microscopic/topological origin of such a defect is deferred. In spherical coordinates (r, θ, ϕ) with the z -axis chosen along the angular momentum $\mathbf{J}$, we define the far-zone *physical* azimuthal speed by

$$v_\phi(r, \theta) = \frac{D}{r^2} \sin \theta, \quad (r \gg a), \quad (19)$$

where D is an effective spin-flow strength fixed by matching the Kerr weak-field gravitomagnetic potential. In these coordinates one should also distinguish physical and covariant azimuthal components: $v_\phi^{(\text{cov})} = r \sin \theta v_\phi^{(\text{phys})}$, so $v_\phi^{(\text{phys})} \propto 1/r^2$ implies $v_\phi^{(\text{cov})} \propto 1/r$, consistent with the Kerr weak-field scaling $g_{0\phi} \propto \sin^2 \theta / r$ and hence $\Omega \propto 1/r^3$.

To connect this flow to gravitomagnetism we use the same acoustic metric dictionary as in the scalar and optical sectors. At leading order in the flow speed, the effective line element for test defects moving in the superfluid can be written schematically as

$$ds^2 = - \left(1 + \frac{2\Phi}{c^2} \right) c^2 dt^2 - \frac{4}{c^3} \mathbf{A}_{\text{eff}} \cdot d\mathbf{x} dt + \left(1 - \frac{2\Psi}{c^2} \right) d\mathbf{x}^2, \quad (20)$$

where Φ and Ψ are the scalar potentials already fixed by Papers I and II, and $\mathbf{A}_{\text{eff}}$ is an effective vector potential proportional to the bulk flow velocity $\mathbf{v}$. For an irrotational flow $\mathbf{A}_{\text{eff}}$ can be removed by a gauge transformation; for the vortical dyon flow in Eq. (19) it carries physical content and directly encodes g_{0i} at 1PN order.

Writing

$$\mathbf{A}_{\text{eff}}(\mathbf{r}) = \kappa \rho_0 \mathbf{v}(\mathbf{r}), \quad (21)$$

with ρ_0 the far-field density and κ a constant of proportionality determined by the underlying hydrodynamics, and inserting Eq. (19) into the off-diagonal part of Eq. (20), one finds a gravitomagnetic potential of the form

$$g_{0i}^{(\text{dyon})} = -\frac{2G}{c^3} \epsilon_{ijk} \frac{\tilde{J}^j x^k}{r^3}, \quad \tilde{\mathbf{J}} = \alpha_D D \hat{\mathbf{z}}, \quad (22)$$

for some dimensionless constant α_D . Matching this to the GR expression fixes the relation between the dyon spin-flow strength D and the physical angular momentum $\mathbf{J}$:

$$D = \frac{4G}{c^2} J, \quad (23)$$

up to the same sign conventions used to orient the circulation and the spin. In other words, once the inertial mass M and spin J of a defect are specified, the strength of its vortex ring is not a new free parameter: it is fixed by the requirement that the far-field gravitomagnetic potential coincide with the 1PN Kerr limit.

3.3 Acoustic metric and observable spin tests

Given the calibration in Eq. (23), the dyon construction reproduces not only the form of g_{0i} but also the standard spin precession observables of GR. The precession of a gyroscope with spin $\mathbf{S}$ at position $\mathbf{r}$ in a stationary spacetime with metric $g_{\mu\nu}$ is governed, to 1PN order, by the equation

$$\frac{d\mathbf{S}}{dt} = \boldsymbol{\Omega} \times \mathbf{S}, \quad (24)$$

where $\boldsymbol{\Omega}$ receives a contribution from the gravitomagnetic potential g_{0i} . Inserting the dyon-induced g_{0i} into the standard PN precession formula yields exactly the Lense–Thirring vector $\boldsymbol{\Omega}_{\text{LT}}(\mathbf{r})$ of Eq. (16) with $\mathbf{J}$ identified as the defect spin.

Similarly, the precession of the orbital plane of a test defect moving in the field of a spinning dyon reproduces the GR nodal precession rate. For a nearly circular orbit of radius r around a central dyon of mass M and spin J aligned with the z -axis, the rate of change of the longitude of the ascending node is

$$\dot{\Omega}_{\text{node}} = \frac{2GJ}{c^2 r^3}, \quad (25)$$

to leading order in J . This matches the classic GR Lense–Thirring result for satellites around the Earth, and is the quantity measured by experiments such as LAGEOS and Gravity Probe B. In the toy universe, once J is specified, this precession rate is an output of the dyon flow; there is no room to independently adjust the strength of the gravitomagnetic coupling.

3.4 Falsifiability in the spin sector

From the standpoint of the toy model, the spin sector introduces no new continuous parameters beyond the physical angular momentum $\mathbf{J}$ of each defect. The mapping between J and the dyon spin-flow strength D in Eq. (23) is fixed by the requirement that the far-field gravitomagnetic potential match the Kerr limit of GR. Given this calibration, the same dyon construction determines:

- the Lense–Thirring precession of gyroscopes in orbit around a spinning mass,
- the nodal precession of orbital planes for test defects,
- and, as we will see in Sec. 4, the vector part of the N -body interaction encoded in the EIH Lagrangian via translational wakes of moving defects.

This rigidity makes the spin sector sharply falsifiable. If future measurements of frame-dragging around rotating bodies were to deviate from the GR Lense–Thirring predictions, the dyon construction in its present form would fail. Conversely, the fact that a single composite defect—a flux-tube sink bound to a vortex ring—can reproduce both the scalar and spin-induced 1PN phenomenology with no additional tuning is a non-trivial consistency check of the superfluid toy universe. In the next section we extend the construction to translational wakes sourced by moving defects to address the full N -body problem and the structure of the EIH Lagrangian.

4 N-Body Dynamics and the EIH Lagrangian

The scalar and optical sectors of the toy universe already reproduce the 1PN tests that probe a single, essentially isolated mass: the Newtonian potential and its scalar correction fix the perihelion advance, while the $n = 5$ refractive index recovers light bending, Shapiro delay, and gravitational

redshift with the GR coefficients. To complete the 1PN picture one must address the dynamics of multiple bodies. In GR this is encoded in the Einstein–Infeld–Hoffmann (EIH) Lagrangian, which describes the relative motion of point masses in the weak-field, slow-motion regime. In this section we show how the same superfluid toy model gives rise to the EIH structure, and what constraints this imposes on the underlying hydrodynamics.

4.1 The EIH target and sector decomposition

For an N -body system of point masses $\{m_A\}$ with positions $\{\mathbf{x}_A\}$ and velocities $\{\mathbf{v}_A\}$, the EIH Lagrangian at 1PN order can be written schematically as

$$L_{\text{EIH}} = L_{\text{N}} + \frac{1}{c^2} L_{\text{1PN}} + \mathcal{O}\left(\frac{1}{c^4}\right), \quad (26)$$

with a Newtonian part

$$L_{\text{N}} = \sum_A \frac{1}{2} m_A v_A^2 + \frac{1}{2} \sum_{A \neq B} \frac{G m_A m_B}{r_{AB}}, \quad (27)$$

and a 1PN correction that splits into three qualitatively distinct pieces:

$$L_{\text{1PN}} = L_{\text{kin}} + L_{\text{stat}} + L_{\text{vec}}. \quad (28)$$

Here L_{kin} is a purely kinetic correction of order v^4 ,

$$L_{\text{kin}} = \sum_A \frac{1}{8} m_A v_A^4, \quad (29)$$

L_{stat} collects the static nonlinear terms proportional to G^2 which couple three masses at a time,

$$L_{\text{stat}} \sim \sum_{A \neq B \neq C} \frac{G^2 m_A m_B m_C}{r_{AB} r_{AC}}, \quad (30)$$

and L_{vec} contains the velocity-dependent pairwise interaction terms. For two bodies A and B , the latter can be written in the form

$$L_{\text{vec}}^{(AB)} = \frac{G m_A m_B}{r_{AB}} \left[\frac{3}{2} (v_A^2 + v_B^2) - \frac{7}{2} \mathbf{v}_A \cdot \mathbf{v}_B - \frac{1}{2} (\mathbf{v}_A \cdot \mathbf{n}_{AB})(\mathbf{v}_B \cdot \mathbf{n}_{AB}) \right], \quad (31)$$

where $\mathbf{n}_{AB} = (\mathbf{x}_A - \mathbf{x}_B)/r_{AB}$ is the unit separation vector. The three coefficients in Eq. (31) are highly constrained: they encode, in a compact way, the vector and tensor structure implied by the underlying metric theory.

In the toy superfluid universe, these three pieces have natural interpretations:

- The v^4 term L_{kin} arises from the relativistic expansion of the defect kinetic energy in the effective metric fixed by Papers I and II.
- The static nonlinear term L_{stat} reflects the fact that the defect mass depends on the local pressure and density; this is essentially the statement that “gravity gravitates” in the scalar sector.
- The velocity-dependent term L_{vec} is generated by the interaction energy of overlapping dyon flows. Its detailed tensor structure depends on how the transverse (vortical) and longitudinal (compressible) components of the flow are coupled.

In what follows we focus on the static nonlinear and vector pieces, which contain the genuinely new physics from the perspective of the toy model. The purely kinetic correction will be assumed to take its standard relativistic form in the emergent metric.

4.2 Static non-linearity (cavitation)

In the superfluid picture, a defect does not carry a rigid, fixed mass independent of its environment. Instead, its effective mass is a property of the throat immersed in the surrounding vacuum: it depends on the local pressure, density, and potential. In Paper I this was encoded in a position-dependent kinetic prefactor $\sigma(r)$, which can be rephrased as a density-dependent mass $m(\rho)$. For a defect labeled A we can write, to leading order in the perturbation of the vacuum,

$$m_A(\mathbf{x}_A) = m_{A,0} \left[1 + \kappa_\rho \frac{\Phi_{\text{loc}}(\mathbf{x}_A)}{c^2} + \mathcal{O}\left(\frac{\Phi^2}{c^4}\right) \right], \quad (32)$$

where $m_{A,0}$ is the bare mass, Φ_{loc} is the local effective potential generated by all other defects and the background vacuum, and κ_ρ is a dimensionless density-depletion coefficient fixed by the cavitation analysis of Paper I. In the calibration used throughout Papers I and II and the present work one has $\kappa_\rho = 1$, which ensures that the resulting static G^2 three-body term matches the Einstein–Infeld–Hoffmann coefficient associated with the PPN parameter $\beta_{\text{PPN}} = 1$. The same cavitation analysis used in Paper I to calibrate the throat aspect ratio L/a also decomposes the inertial parameter $\beta = 3$ into density and pressure–volume contributions, with κ_ρ capturing the density-depletion piece.

To see how this generates the static G^2 term, consider the Newtonian potential energy between bodies A and B ,

$$V_{AB}^{\text{N}} = -\frac{G m_A(\mathbf{x}_A) m_B(\mathbf{x}_B)}{r_{AB}}. \quad (33)$$

Inserting Eq. (32) for each mass and expanding to first order in Φ_{loc}/c^2 produces correction terms of order G^2/c^2 . For instance, the mass of body A picks up a contribution from the potential generated by a third body C ,

$$\Phi_{\text{loc}}(\mathbf{x}_A) \supset -\frac{G m_C}{r_{AC}}, \quad (34)$$

so that the AB interaction energy acquires a correction

$$\delta V_{AB}^{(C)} = -\frac{G}{r_{AB}} \left[\kappa_\rho \frac{m_{A,0} m_{B,0}}{c^2} \Phi_{\text{loc}}(\mathbf{x}_A) + (A \leftrightarrow B) \right] \supset \kappa_\rho \frac{G^2 m_{A,0} m_{B,0} m_C}{c^2 r_{AB} r_{AC}}. \quad (35)$$

Summing over all triplets (A, B, C) and symmetrizing produces a three-body interaction energy of the schematic form

$$V_{\text{stat}}^{(3)} = - \sum_{A \neq B \neq C} \frac{G^2 m_A m_B m_C}{c^2} F_{\text{stat}}(r_{AB}, r_{AC}, r_{BC}), \quad (36)$$

with

$$F_{\text{stat}}(r_{AB}, r_{AC}, r_{BC}) \propto \frac{1}{r_{AB} r_{AC}} + \text{permutations}. \quad (37)$$

The precise numerical coefficient of this term is determined by κ_ρ and the details of how Φ_{loc} is assembled from the scalar and optical sectors; the Mathematica analysis shows that, with $\kappa_\rho = 1$ fixed by the single-body cavitation analysis, the resulting three-body term agrees with the static G^2 part of the EIH Lagrangian.

Conceptually, this mechanism is nothing but “cavitation” in the vacuum: defects are holes that displace fluid, and the amount of fluid displaced depends on the local pressure and potential, which in turn are sourced by other defects. Gravity gravitates because defects feel not only the potential, but also the potential of the potential, through their density-dependent mass.

4.3 Vector interaction: translational wakes

We now turn to the velocity-dependent *cross* terms in the Einstein–Infeld–Hoffmann (EIH) Lagrangian, i.e. the coefficients of $\mathbf{v}_A \cdot \mathbf{v}_B$ and $(\mathbf{v}_A \cdot \mathbf{n}_{AB})(\mathbf{v}_B \cdot \mathbf{n}_{AB})$ in Eq. (31). In the toy universe these terms arise from the overlap energy of the *translational wakes* sourced by *moving* defects.

It is crucial to distinguish this translation-driven wake from the rotation-driven flow used in the spin sector (Sec. 3). A spinning defect sources a vortical far field calibrated by matching to Lense–Thirring (J/r^3), whereas the EIH cross terms depend on linear velocities $\mathbf{v}_A, \mathbf{v}_B$ and must be sourced by the defect’s *translational* motion.

We retain the overlap-energy definition

$$V_{\text{vec}}^{(AB)} = \rho_0 \int d^3x \mathbf{u}_A(\mathbf{x}) \cdot \mathbf{u}_B(\mathbf{x}), \quad (38)$$

but now interpret $\mathbf{u}_{A,B}$ as translational wake fields. To obtain a pairwise interaction that falls off as $1/r_{AB}$ at 1PN order, the wake must scale as $u(\mathbf{k}) \propto 1/k$ in Fourier space so that $u_A \cdot u_B \propto 1/k^2$.

We therefore parametrize the most general isotropic linear-response wake as a sum of transverse and longitudinal projector pieces (and an optional helical transverse mode):

$$\mathcal{P}_L^{ij}(\mathbf{k}) = \frac{k^i k^j}{k^2}, \quad \mathcal{P}_T^{ij}(\mathbf{k}) = \delta^{ij} - \frac{k^i k^j}{k^2}, \quad (39)$$

$$\mathbf{u}_{\text{trans}}(\mathbf{k}; \mathbf{v}) = i \frac{K}{k} \mathcal{P}_T(\mathbf{k}) \mathbf{v} + i \frac{K a_H}{k^2} (\mathbf{k} \times \mathbf{v}) + i \frac{K \alpha}{k^3} \mathbf{k} (\mathbf{k} \cdot \mathbf{v}), \quad (40)$$

where K is an overall coupling, α controls the longitudinal (compressible) wake admixture, and a_H parametrizes a helical transverse component (set to zero in the minimal EIH match).

Evaluating the overlap integral with the wake ansatz (40) produces a $1/r_{AB}$ pair interaction of the EIH form,

$$V_{\text{vec}}^{(AB)} = \frac{G m_A m_B}{c^2 r_{AB}} [C_{\parallel}(\alpha, a_H) \mathbf{v}_A \cdot \mathbf{v}_B + C_L(\alpha, a_H) (\mathbf{v}_A \cdot \mathbf{n}_{AB})(\mathbf{v}_B \cdot \mathbf{n}_{AB}) + \dots], \quad (41)$$

where the ellipsis denotes terms that renormalize into the self-velocity channel and/or the already-calibrated kinetic prefactors. The nontrivial information for the EIH tensor match lies in the cross-term coefficients $C_{\parallel}$ and C_L , derived in Sec. 4.4 and Appendix C.

4.4 Derivation walkthrough: matching the EIH cross tensor

Using the translational wake ansatz (40) in the overlap integral (38), the EIH cross-term coefficients take the universal form (Appendix C)

$$C_{\parallel}(\alpha, a_H) = K \pi^2 (-1 + a_H^2 - \alpha^2), \quad (42)$$

$$C_L(\alpha, a_H) = K \pi^2 (-1 + a_H^2 + \alpha^2), \quad (43)$$

up to the overall coupling K .

The EIH targets for the cross terms in Eq. (31) are

$$C_{\parallel}^{\text{EIH}} = -\frac{7}{2}, \quad C_L^{\text{EIH}} = -\frac{1}{2}. \quad (44)$$

Taking the ratio eliminates K and yields

$$\frac{C_L}{C_{\parallel}} = \frac{-1 + a_H^2 + \alpha^2}{-1 + a_H^2 - \alpha^2} = \frac{1}{7}, \quad (45)$$

which implies

$$\alpha^2 = \frac{3}{4}(1 - a_H^2). \quad (46)$$

The minimal (least structured) wake match sets $a_H = 0$, giving the real value

$$\alpha^2 = \frac{3}{4}. \quad (47)$$

With this choice, the overall coupling is fixed by the parallel coefficient (42):

$$K = \frac{2}{\pi^2}. \quad (48)$$

Together these reproduce the EIH cross coefficients exactly:

$$C_{\parallel} = -\frac{7}{2}, \quad C_L = -\frac{1}{2}. \quad (49)$$

In particular, no imaginary compressibility parameter is required. The earlier appearance of $\alpha^2 < 0$ was an artifact of restricting the translation wake to an incomplete transverse basis; once the full isotropic projector decomposition is used, the EIH tensor is obtained with real parameters. The coefficient forms and tuning are verified symbolically in the accompanying Mathematica script `mathematica/n_body.wl`.

4.5 Stability and interpretation of the wake parameters

The matching in Sec. 4.4 fixes the relative weight of the longitudinal (compressible) and transverse (solenoidal/vortical) components of the *translational* wake.¹ The required value $\alpha^2 = 3/4$ is real, so the corresponding quadratic wake functional is positive-definite in the usual Euclidean hydrodynamic sense.

Equally importantly, the minimal match sets the optional helical transverse component to zero ($a_H = 0$). This cleanly separates the sectors of the toy model: spin observables are governed by the rotational/vortical flow calibrated in Sec. 3, whereas the EIH velocity-dependent cross terms are governed by the translational wake response.

The remaining physical hypothesis is dynamical rather than algebraic: the medium must support a long-range translational wake with Fourier scaling $u(\mathbf{k}) \propto 1/k$ (equivalently, $u(\mathbf{x}) \propto 1/r^2$) so that the overlap energy produces a $1/r$ interaction at 1PN order. Unlike the $1/r^3$ dipolar disturbance produced by a translating impermeable rigid inclusion in potential flow (schematically $\delta\mathbf{v} \sim a^3\mathbf{U}/r^3$), this $1/r^2$ scaling is consistent with a defect that continuously exchanges mass/flux with the vacuum—an effective sink, i.e. a monopole-type far-field flow in the effective 3D description. In the throat picture, this corresponds to exchange with a bulk reservoir rather than literal mass nonconservation in a closed fluid. Whether a stiff vacuum equation of state (such as the $n = 5$ polytrope inferred in Paper II [2]) naturally generates this response is a question for the PDE-level simulations. In the present work we show that *if* such wakes exist, their tensor structure can reproduce the EIH interaction with real, stable parameters.

¹Here “transverse” is meant in the Fourier–projector sense, $\mathbf{k} \cdot \mathbf{u}_T = 0$: a solenoidal (vortical) wake component sourced by defect circulation, not an elastic shear modulus.

5 Discussion and Outlook

5.1 Summary: 1PN completion

Taken together, the three papers in this series show how a single, minimal superfluid–defect toy model can reproduce the full suite of 1PN solar-system tests usually attributed to the Schwarzschild and Kerr solutions of GR with PPN parameters $\beta_{\text{PPN}} = \gamma_{\text{PPN}} = 1$. It is useful to summarize the structure of the construction and the status of the free parameters.

In Paper I [1], the focus was on orbital dynamics in the static, spherically symmetric sector. A scalar lag field and a position-dependent kinetic prefactor $\sigma(r)$ were introduced to model how defects move relative to the bulk vacuum and how their inertia is renormalized by the throat background. In the static single-body limit the lag field relaxes, so the 1PN perihelion advance is generated entirely by the inertia profile $\sigma(r)$. By demanding that nearly Keplerian orbits around an isolated defect reproduce the observed 1PN perihelion advance, the analysis fixed a single dimensionless parameter $\beta = 3$ and constrains the throat aspect ratio L/a through the pressure–volume coefficient. The same cavitation analysis ties the defect mass to the surrounding density and selects $\kappa_\rho = 1$, which sets the strength of the static G^2 term in the N -body problem. Within this calibration, the scalar sector of the model yields an effective g_{00} component that matches GR at 1PN order.

Paper II [2] extended the same toy universe to the optical and clock sectors. Treating the vacuum as a stiff ($n = 5$) polytropic superfluid endowed the medium with a density-dependent sound speed and hence a radial refractive index profile $N(r)$ around a defect. By constructing an effective optical metric from $N(r)$ and comparing to the Schwarzschild metric, the analysis showed that $n = 5$ is uniquely selected (within the class considered) by the requirements of matching 1PN light bending, Shapiro time delay, and gravitational redshift. The resulting effective spacetime has PPN parameters $\beta_{\text{PPN}} = \gamma_{\text{PPN}} = 1$ once the scalar and optical contributions are combined, so the toy model reproduces both the orbital and optical 1PN tests with no new free parameters beyond those already fixed in Paper I.

The present work adds the remaining ingredients: spin-induced gravitomagnetism and the full N -body structure encoded in the Einstein–Infeld–Hoffmann Lagrangian. By promoting defects to composite dyons (flux-tube sinks bound to vortex rings), we showed that the far-field vorticity reproduces the Lense–Thirring gravitomagnetic potential with the correct J/r^3 scaling, fixing the relation between vortex strength and angular momentum J by matching to the Kerr weak-field limit. Separately, the velocity-dependent *cross* terms in the EIH Lagrangian are reproduced by the overlap energy of translational wakes sourced by moving defects. Using the isotropic transverse/longitudinal projector decomposition, the EIH cross-term tensor fixes a real wake mixing $\alpha^2 = 3/4$ (with minimal $a_H = 0$) and an overall coupling $K = 2/\pi^2$.

With this choice, the 1PN dynamics of the toy universe—including single-body orbits, light propagation, spin precession, and general N -body motion—are fully specified by a small, tightly constrained set of parameters:

- the throat geometry (a fixed aspect ratio L/a),
- the scalar renormalization parameter ($\beta = 3$),
- the stiff polytropic index ($n = 5$),
- and the dyon spin calibration ($D \propto GJ/c^2$) together with the translational wake tuning ($a_H = 0$, $\alpha^2 = 3/4$, and $K = 2/\pi^2$).

No further continuous freedom remains at 1PN order within this framework.

5.2 Emergent metric and effective field theory viewpoint

Although the toy universe is formulated in hydrodynamic language, it is best understood as an emergent metric theory. The mapping can be summarized schematically as follows:

- The *defects* (flux-tube throats) and their geometry encode the rest masses of gravitating bodies and fix the scale $\mu = GM$ that appears in the effective potentials.
- The scalar lag field and density deficit determine $\Phi(r)$ and the inertial renormalization $\sigma(r)$, and thereby fix the effective g_{00} component relevant for slow-motion dynamics.
- The refractive index profile $N(r)$ of the stiff $n = 5$ vacuum captures the optical manifestations of the spatial metric, leading to an effective g_{ij} with $\gamma = 1$ in the PPN expansion.
- The bulk flow and vorticity generated by spinning dyons furnish an effective vector potential and hence the off-diagonal g_{0i} components responsible for gravitomagnetism and the vector part of the EIH interaction.

In this sense the superfluid variables $(\rho, p, \mathbf{v})$ provide a particular parameterization of the metric and its first PN corrections, constrained by the throat microphysics and the equation of state.

From an effective field theory (EFT) perspective, what the present series accomplishes is a highly nontrivial matching. A generic metric EFT with a symmetry structure comparable to GR contains many a priori independent coefficients at 1PN order, which are usually encoded in the PPN parameters and higher-derivative operators. Here, by contrast, all of the 1PN coefficients are determined by a handful of hydrodynamic response parameters of a single medium. The fact that these parameters can be chosen once and for all to reproduce the GR phenomenology suggests that the superfluid-defect picture captures a nontrivial subset of metric EFTs. At the same time, the requirement that the scalar ($q = 1$) [1] and stiff optical ($n = 5$) [2] sectors dovetail with the translational wake tuning ($a_H = 0$, $\alpha^2 = 3/4$, $K = 2/\pi^2$) indicates that not every classical fluid admits such an interpretation: the vacuum thermodynamics and linear response are tightly constrained.

5.3 Limitations and open questions

The superfluid toy universe is deliberately minimal, and its domain of applicability is correspondingly limited. Several caveats and open questions are worth highlighting.

First, the analysis is restricted to the weak-field, slow-motion regime in which a 1PN expansion is valid. We have not attempted to extend the construction to strong-field situations where horizons, ergoregions, or large curvature effects are important. Whether the same superfluid medium can support analogues of black holes or compact binaries that reproduce GR predictions beyond 1PN is an open question.

Second, radiation and dissipation have been neglected. The present treatment describes conservative dynamics: there is no gravitational-wave emission, no radiation reaction, and no associated energy loss from the system. In a genuine emergent-gravity scenario one would expect the superfluid to support wave excitations that play the role of gravitational radiation, and the dyon dynamics would have to be augmented to account for their backreaction. Developing a wave sector consistent with the 1PN matching presented here is an important next step.

Third, the internal structure of the defects has been treated in a highly compressed way. Throat geometry enters only through a small number of coarse-grained parameters (a , L , and the circulation Γ), and the dyon parameters (α , a_H , K) summarize the longitudinal/transverse/helical translational

wake response. A more microscopic model of the throat—for example, one rooted in a condensed-matter analogue or in a specific quantum field theory of the vacuum—could clarify whether the required thermodynamic choices ($q = 1$, $n = 5$) [1, 2] and wake tuning ($a_H = 0$, $\alpha^2 = 3/4$, $K = 2/\pi^2$) arise naturally or must be imposed by hand. Future work will investigate whether these specific values—especially $n = 5$ and $\alpha^2 = 3/4$ —arise from throat boundary conditions in a higher-dimensional (brane–bulk) construction, with defect mass and spin expressed as flux integrals of the underlying geometry.

Fourth, we have so far considered only uncharged, purely gravitational defects. The electromagnetic sector is conspicuously absent. In a fully unified analogue model, electric charge and the Maxwell field would emerge as additional collective modes of the same vacuum medium, possibly tied to different components of the throat or its winding structure. How the EM sector couples to the superfluid gravity described here, and whether it preserves the successful 1PN matching, remains to be seen.

Finally, the connection to cosmology has been left unexplored. The present work treats the vacuum medium as homogeneous and static on large scales, with no global expansion or background flow. Embedding the superfluid-defect construction into an expanding cosmological background would raise new questions about the role of defects in structure formation, the effective cosmological constant, and possible departures from GR on large scales.

5.4 Observational handles

Although the toy universe is not intended as a replacement for GR, it does offer a well-defined set of observational handles. By construction, any deviation from the 1PN phenomenology of GR in the regimes considered here would falsify the specific parameter choices that underlie the model.

In the scalar and optical sectors, the primary tests are the classic solar-system measurements: perihelion precession, light deflection by the Sun, Shapiro time delay, and gravitational redshift. These observables were used to fix β , n , and L/a , so they serve more as consistency checks than as independent predictions. Nevertheless, any future refinement that revealed tension among these constraints would feed back into the allowed parameter space of the superfluid description.

In the spin sector, frame-dragging measurements provide a sharper probe. Experiments such as Gravity Probe B and the analysis of LAGEOS satellite orbits already constrain the Lense–Thirring precession around the Earth to within a few tens of percent. Within the toy model, the same calibration that matches Kerr at 1PN fixes the relation between spin J and vortex strength D , so the frame-dragging rates of all spinning defects are tightly linked. Any confirmed discrepancy between observed and GR-predicted frame-dragging in the weak-field regime would either rule out the dyon construction or force a revision of the underlying hydrodynamic dictionary.

In the N -body sector, the EIH Lagrangian governs the dynamics of weakly bound systems ranging from planetary orbits to wide binaries. Precision ephemerides in the solar system and timing observations of pulsar binaries already test the EIH tensor structure to high accuracy. Since the superfluid model reproduces this structure only when the scalar ($q = 1$) [1] and optical ($n = 5$) [2] sectors dovetail with the translational wake tuning ($a_H = 0$, $\alpha^2 = 3/4$, $K = 2/\pi^2$), improved measurements of velocity-dependent effects in multi-body systems can be interpreted as tests of this specific hydrodynamic response.

Looking ahead, the most discriminating tests are likely to arise beyond the strict 1PN regime: in systems where 2PN corrections, spin couplings, and radiation reaction all play important roles. Extending the superfluid-defect toy universe into those domains would either uncover qualitative departures from GR—providing concrete targets for observation—or further cement the correspondence between hydrodynamic and geometric descriptions of gravity. In either case, the present 1PN

completion offers a useful baseline: it demonstrates that, at least in principle, a remarkably small amount of structured “fluid” can mimic the familiar phenomenology of curved spacetime.

A Dyon far-field flow and gravitomagnetic calibration

This appendix records only an effective far-field definition and calibration of the dyon spin flow used in Sec. 3. In this paper we define the dyon’s far-zone behavior by matching to the GR weak-field Kerr/Lense–Thirring sector; the microscopic or topological origin of such a defect is deferred.

A.1 GR target

The weak-field Kerr gravitomagnetic potential can be written as

$$g_{0i}^{(\text{GR})} = -\frac{2G}{c^3} \epsilon_{ijk} \frac{J^j x^k}{r^3}, \quad (50)$$

which implies the familiar spherical-component scaling $g_{0\phi} \propto \sin^2 \theta / r$ and hence $\mathbf{\Omega}_{\text{LT}} \propto 1/r^3$.

A.2 Acoustic mapping + required flow

As in the main text, we take the off-diagonal metric term to be sourced by an effective vector potential $\mathbf{A}_{\text{eff}}$ proportional to the bulk flow $\mathbf{v}$,

$$ds^2 \supset -\frac{4}{c^3} \mathbf{A}_{\text{eff}} \cdot d\mathbf{x} dt, \quad \mathbf{A}_{\text{eff}}(\mathbf{r}) = \kappa \rho_0 \mathbf{v}(\mathbf{r}), \quad (51)$$

with ρ_0 the far-field density and κ a constant fixed by the hydrodynamic dictionary used throughout the paper.

The phenomenological far-zone azimuthal flow used for the dyon is

$$v_\phi(r, \theta) = \frac{D}{r^2} \sin \theta, \quad (52)$$

where v_ϕ is the *physical* azimuthal speed and D is an effective spin-flow strength. In spherical coordinates the corresponding covariant component carries an additional factor,

$$v_\phi^{(\text{cov})} = r \sin \theta v_\phi^{(\text{phys})}, \quad (53)$$

so $v_\phi^{(\text{phys})} \propto 1/r^2$ implies $v_\phi^{(\text{cov})} \propto 1/r$, consistent with the Kerr scaling $g_{0\phi} \propto \sin^2 \theta / r$.

A.3 Calibration

With the above dictionary, matching the dyon-induced g_{0i} to the GR target (50) fixes the same spin calibration used in the main text,

$$D = \frac{4G}{c^2} J, \quad (54)$$

so the dyon spin-flow strength is not a new free parameter once the physical spin J is specified. This appendix therefore serves only to state the effective far-field definition and calibration; deeper microscopic explanations are left to future work.

B Static non-linearity derivation

In this appendix we make the derivation in Sec. 4.2 explicit. The goal is to show how a density-dependent defect mass $m_A(\rho)$ generates a three-body static term of order G^2 with the characteristic EIH structure

$$L_{\text{stat}} \sim \sum_{A \neq B \neq C} \frac{G^2 m_A m_B m_C}{c^2 r_{AB} r_{AC}}, \quad (55)$$

up to a numerical coefficient fixed by the same pressure–volume analysis used in Paper I to calibrate L/a and to decompose the inertial parameter $\beta = 3$ into density and pressure–volume contributions.

B.1 Density-dependent mass and local potential

The key physical input is that the effective mass of a defect is not a fixed constant but depends on the local properties of the vacuum medium. In the coarse-grained description this dependence can be parametrized by the local effective potential $\Phi_{\text{loc}}(\mathbf{x})$, which encodes both the scalar lag field and the pressure–volume response of the throat. To leading order one may write

$$m_A(\mathbf{x}_A) = m_{A,0} \left[1 + \kappa_\rho \frac{\Phi_{\text{loc}}(\mathbf{x}_A)}{c^2} + \mathcal{O}\left(\frac{\Phi^2}{c^4}\right) \right], \quad (56)$$

where $m_{A,0}$ is the bare mass parameter of defect A , $\Phi_{\text{loc}}(\mathbf{x}_A)$ is the effective potential at the defect location, and κ_ρ is a dimensionless density–response coefficient which, in the calibration of Paper I, satisfies $\kappa_\rho = 1$. In the main text this is Eq. (32).

For a configuration of N defects, the local potential can be split into contributions from individual sources,

$$\Phi_{\text{loc}}(\mathbf{x}_A) = \sum_{B \neq A} \Phi_B(\mathbf{x}_A) + \Phi_{\text{vac}}(\mathbf{x}_A), \quad (57)$$

where Φ_B is the field due to defect B and Φ_{vac} is the (slowly varying) background contribution. At the level of the static G^2 term we may drop the background, since it does not generate the $1/r_{AB} r_{AC}$ structure of interest. In the weak-field regime the field of an isolated defect is

$$\Phi_B(\mathbf{x}) = -\frac{G m_{B,0}}{|\mathbf{x} - \mathbf{x}_B|} + \mathcal{O}\left(\frac{G^2 m_{B,0}^2}{c^2 r^2}\right), \quad (58)$$

so that, to Newtonian accuracy,

$$\Phi_{\text{loc}}(\mathbf{x}_A) \simeq -\sum_{C \neq A} \frac{G m_{C,0}}{r_{AC}}, \quad r_{AC} = |\mathbf{x}_A - \mathbf{x}_C|. \quad (59)$$

Inserting Eq. (59) into Eq. (56) and keeping terms up to $\mathcal{O}(G/c^2)$ gives

$$m_A(\mathbf{x}_A) = m_{A,0} - \kappa_\rho \frac{G m_{A,0}}{c^2} \sum_{C \neq A} \frac{m_{C,0}}{r_{AC}} + \mathcal{O}\left(\frac{G^2}{c^4}\right). \quad (60)$$

B.2 Newtonian pair potential with dressed masses

The Newtonian potential energy of an N -body configuration with positions $\{\mathbf{x}_A\}$ and effective masses $\{m_A\}$ is

$$V_N = -\frac{1}{2} \sum_{A \neq B} \frac{G m_A(\mathbf{x}_A) m_B(\mathbf{x}_B)}{r_{AB}}, \quad (61)$$

where the factor of $1/2$ avoids double counting of pairs. Substituting Eq. (60) for each mass and expanding to first order in $\kappa_\rho G/c^2$ yields

$$V_N = -\frac{1}{2} \sum_{A \neq B} \frac{G}{r_{AB}} \left\{ m_{A,0} m_{B,0} - \kappa_\rho \frac{G m_{A,0} m_{B,0}}{c^2} \sum_{C \neq A} \frac{m_{C,0}}{r_{AC}} - \kappa_\rho \frac{G m_{A,0} m_{B,0}}{c^2} \sum_{D \neq B} \frac{m_{D,0}}{r_{BD}} + \mathcal{O}\left(\frac{G^2}{c^4}\right) \right\}. \quad (62)$$

The first term in braces reproduces the usual Newtonian potential energy,

$$V_N^{(0)} = -\frac{1}{2} \sum_{A \neq B} \frac{G m_{A,0} m_{B,0}}{r_{AB}}. \quad (63)$$

The remaining terms are of order G^2/c^2 and generate the static three-body interaction that we wish to isolate.

It is convenient to focus on the correction associated with a particular triple (A, B, C) . Consider the piece of V_N in which the mass of A is dressed by the potential of C :

$$\delta V_{AB}^{(C)} = -\frac{1}{2} \sum_{A \neq B} \frac{G}{r_{AB}} \left[-\kappa_\rho \frac{G m_{A,0} m_{B,0}}{c^2} \sum_{C \neq A} \frac{m_{C,0}}{r_{AC}} \right]. \quad (64)$$

Extracting the contribution with a specific C and suppressing the subscript 0 on the bare masses for readability, we can write

$$\delta V_{AB}^{(C)} = \frac{\kappa_\rho G^2}{2c^2} \sum_{A \neq B} \sum_{C \neq A} \frac{m_A m_B m_C}{r_{AB} r_{AC}}. \quad (65)$$

A completely analogous term arises from dressing the mass of B by the potential of a defect D ,

$$\delta V_{BA}^{(D)} = \frac{\kappa_\rho G^2}{2c^2} \sum_{A \neq B} \sum_{D \neq B} \frac{m_A m_B m_D}{r_{AB} r_{BD}}. \quad (66)$$

B.3 Symmetrization over triples and EIH form

The sum in Eq. (65) runs over all ordered pairs (A, B) and, for each A , over all $C \neq A$. To make contact with the usual EIH notation it is helpful to rewrite the result as a fully symmetric sum over unordered triples. Define the triple sum

$$\sum'_{A,B,C} \equiv \sum_{\substack{A,B,C \\ \text{all distinct}}} , \quad (67)$$

so that each unordered set $\{A, B, C\}$ appears $3! = 6$ times in the primed sum. Then the combined three-body correction from dressing both A and B can be written schematically as

$$V_{\text{stat}}^{(3)} = \frac{\kappa_\rho G^2}{2c^2} \sum'_{A,B,C} m_A m_B m_C \left(\frac{1}{r_{AB} r_{AC}} + \frac{1}{r_{AB} r_{BC}} \right), \quad (68)$$

where the two terms in parentheses correspond to the two ways in which the potential of the third body can dress the masses in the pair. Since the primed sum is fully symmetric in (A, B, C) , the structure in Eq. (68) can be reorganized into the more familiar EIH form

$$V_{\text{stat}}^{(3)} = -C_{\text{stat}} \frac{G^2}{c^2} \sum_{A,B,C}^{\prime} \frac{m_A m_B m_C}{r_{AB} r_{AC}}, \quad (69)$$

for some positive coefficient C_{stat} proportional to κ_ρ . The overall minus sign reflects the convention that V_N is negative for an attractive interaction.

In the calibration used in Papers I and II, the density–depletion coefficient κ_ρ is fixed to unity by the cavitation analysis of a single defect in the background vacuum. With this choice, and including additional scalar contributions subleading in the single-body analysis but of the same order in G^2/c^2 , the Mathematica implementation of the full scalar sector confirms that the three-body term above matches exactly the static G^2 piece of the Einstein–Infeld–Hoffmann Lagrangian, i.e. the “gravity gravitates” term controlled by the PPN parameter $\beta_{\text{PPN}} = 1$.

B.4 Interpretation

From the hydrodynamic perspective, the derivation above can be summarized in a simple slogan: *gravity gravitates because defects are cavities whose mass depends on the ambient potential*. Each defect displaces a volume of vacuum whose mass content is modulated by the local pressure and density. When one defect sits in the field of others, the amount of vacuum it displaces—and hence its effective inertial/gravitational mass—is altered. This dependence shows up as a correction to the pairwise Newtonian potential energy that couples three masses at a time and falls off as $1/(r_{AB} r_{AC})$, just as required by the static part of the EIH Lagrangian.

Crucially, the coefficient of this term is not an independent parameter: it is fixed by the same pressure–volume response of the throat that was already constrained by the single-body 1PN analysis. Once the scalar sector is calibrated to reproduce the perihelion advance and the pressure–volume coefficient, the static G^2 three-body interaction is an unavoidable consequence of the model.

C Vector kernel and α tuning

In this appendix we derive the translational wake overlap coefficients used in Sec. 4.3 and show how the EIH cross-term tensor fixes the wake parameters to real values.

C.1 Fourier-space representation of the translational wake

We assume that a moving defect sources a long-range translational wake whose linear response can be decomposed into transverse and longitudinal projector components (and, optionally, a helical transverse component). With $\mathcal{P}_T^{ij} = \delta^{ij} - k^i k^j / k^2$ and $\mathcal{P}_L^{ij} = k^i k^j / k^2$, we write

$$\mathbf{u}_{\text{trans}}(\mathbf{k}; \mathbf{v}) = i \frac{K}{k} \mathcal{P}_T(\mathbf{k}) \mathbf{v} + i \frac{K a_H}{k^2} (\mathbf{k} \times \mathbf{v}) + i \frac{K \alpha}{k^3} \mathbf{k} (\mathbf{k} \cdot \mathbf{v}), \quad (70)$$

which is Eq. (40) in the main text.

C.2 Overlap integral and coefficient extraction

For two bodies A and B , the interaction energy is

$$V_{\text{vec}}^{(AB)} = \rho_0 \int d^3x \mathbf{u}_A(\mathbf{x}) \cdot \mathbf{u}_B(\mathbf{x}) = \rho_0 \int \frac{d^3k}{(2\pi)^3} \mathbf{u}_A(-\mathbf{k}) \cdot \mathbf{u}_B(\mathbf{k}) e^{i\mathbf{k} \cdot \mathbf{r}_{AB}}. \quad (71)$$

Performing the angular integrals yields a $1/r_{AB}$ interaction of the form (41) with cross-term coefficients

$$C_{\parallel}(\alpha, a_H) = K \pi^2 (-1 + a_H^2 - \alpha^2), \quad (72)$$

$$C_{\text{L}}(\alpha, a_H) = K \pi^2 (-1 + a_H^2 + \alpha^2). \quad (73)$$

C.3 EIH matching and tuned parameters

Matching the EIH cross-term coefficients in Eq. (31) requires

$$C_{\parallel} = -\frac{7}{2}, \quad C_{\text{L}} = -\frac{1}{2}. \quad (74)$$

The ratio constraint gives

$$\alpha^2 = \frac{3}{4} (1 - a_H^2), \quad (75)$$

and the minimal match sets $a_H = 0$, yielding $\alpha^2 = 3/4$ (Eq. (47)). The overall coupling is then fixed by $C_{\parallel} = -7/2$ to

$$K = \frac{2}{\pi^2}, \quad (76)$$

as in Eq. (48). With these choices the wake overlap reproduces the EIH cross tensor exactly.

C.4 Summary of the tuning procedure

For clarity, the translational-wake tuning proceeds as follows:

1. Write the isotropic wake response in projector form (Eq. (40)), with longitudinal mixing parameter α and optional helical amplitude a_H .
2. Evaluate the overlap integral to obtain $C_{\parallel}(\alpha, a_H)$ and $C_{\text{L}}(\alpha, a_H)$.
3. Impose the EIH cross-term targets $(-7/2, -1/2)$ to fix α^2 and then fix the overall coupling K .

D EIH–metric mapping

In this appendix we briefly summarize how the effective metric of the superfluid toy universe maps onto the Einstein–Infeld–Hoffmann (EIH) Lagrangian, and how the parameter choices made across Papers I and II and the present work ensure consistency with the standard post-Newtonian (PN) expansion of GR. The goal is not to reproduce the full EIH derivation, but to show how the scalar, optical, and vector sectors combine into a consistent set of effective 1PN metric data (an effective PN metric for the calibrated dynamics), even though the underlying acoustic (optical) and hydrodynamic (matter) projections need not coincide (bi-metric structure).

D.1 Metric ansatz and PN expansion

In the weak-field, slow-motion regime, it is convenient to write the metric in the usual PN form

$$g_{00} = -1 + \frac{2U}{c^2} - \frac{2\beta_{\text{PPN}}U^2}{c^4} + \mathcal{O}\left(\frac{1}{c^6}\right), \quad (77)$$

$$g_{0i} = -\frac{4V_i}{c^3} + \mathcal{O}\left(\frac{1}{c^5}\right), \quad (78)$$

$$g_{ij} = \left(1 + \frac{2\gamma_{\text{PPN}}U}{c^2}\right) \delta_{ij} + \mathcal{O}\left(\frac{1}{c^4}\right), \quad (79)$$

where U is the Newtonian potential and V_i is the gravitomagnetic vector potential generated by moving and spinning masses. The constants β_{PPN} and γ_{PPN} are the standard PPN parameters; in GR one has $\beta_{\text{PPN}} = \gamma_{\text{PPN}} = 1$.

In the superfluid defect model, the effective metric arises from the scalar potential Φ and inertia profile $\sigma(r)$ (Papers I), the refractive index $N(r)$ of the $n = 5$ vacuum (Paper II), and the dyon flow $\mathbf{v}(\mathbf{x})$ (this paper). Schematically, the mapping can be written as

$$g_{00} = -\left[1 + 2\frac{\Phi_{\text{eff}}}{c^2} + \mathcal{F}(\Phi_{\text{eff}}^2) + \dots\right], \quad (80)$$

$$g_{0i} = -\frac{4}{c^3}A_{\text{eff},i}(\mathbf{x}), \quad (81)$$

$$g_{ij} = \left[1 - 2\frac{\Psi_{\text{eff}}}{c^2}\right] \delta_{ij} + \dots, \quad (82)$$

where Φ_{eff} is the total scalar potential including lag corrections, Ψ_{eff} encodes the spatial curvature inherited from the $n = 5$ vacuum, and $\mathbf{A}_{\text{eff}}$ is proportional to the bulk flow velocity $\mathbf{v}$ of the superfluid around dyons.

Matching to Eqs. (77)–(79) identifies

$$U \equiv -\Phi_{\text{eff}}, \quad (83)$$

$$V_i \equiv A_{\text{eff},i}, \quad (84)$$

and expresses the effective PPN parameters β_{PPN} and γ_{PPN} in terms of the scalar and optical response functions. Paper II showed explicitly that, once the scalar lag and $n = 5$ refractive index are combined with the pressure–volume constraint from Paper I, one obtains

$$\beta_{\text{PPN}} = 1, \quad \gamma_{\text{PPN}} = 1. \quad (85)$$

(Here γ_{PPN} denotes the value inferred from optical observables after applying the dressing relation.) The present work extends this dictionary to include g_{0i} via the dyon flow and thereby fixes the vector potential V_i that appears in the EIH Lagrangian.

D.2 From metric to EIH Lagrangian

Given a metric of the form Eqs. (77)–(79), the EIH Lagrangian follows from expanding the point-particle action

$$S = -\sum_A m_{Ac} \int \sqrt{-g_{\mu\nu} \frac{dx_A^\mu}{dt} \frac{dx_A^\nu}{dt}} dt, \quad (86)$$

in powers of $1/c$ up to $\mathcal{O}(1/c^2)$. The result can be written in the schematic form

$$L_{\text{EIH}} = \sum_A \frac{1}{2} m_A v_A^2 + \frac{1}{2} \sum_{A \neq B} \frac{G m_A m_B}{r_{AB}} + \frac{1}{c^2} L_{\text{1PN}}[U, V_i] + \mathcal{O}\left(\frac{1}{c^4}\right), \quad (87)$$

where L_{1PN} contains the v^4 kinetic term, the static G^2 three-body term, and the velocity-dependent pairwise interactions. The detailed form of L_{1PN} is determined entirely by U , V_i , and the PPN parameters.

Using the standard PN bookkeeping, one finds that:

- The v^4 kinetic term and the $v^2 U$ terms come from expanding $\sqrt{-g_{00} - 2g_{0i}v_A^i/c - g_{ij}v_A^i v_A^j/c^2}$ in powers of v_A/c .
- The static G^2 term arises from the quadratic dependence of U on the masses (“gravity gravitates”) together with the U^2 contribution to g_{00} proportional to β .
- The velocity-dependent pairwise terms come from the cross couplings $g_{0i}v_A^i$ and from the spatial metric g_{ij} , and their tensor structure is governed by the combination of U , V_i , and γ .

For $\beta_{\text{PPN}} = \gamma_{\text{PPN}} = 1$ and for U and V_i satisfying the usual Poisson equations

$$\nabla^2 U = -4\pi G \sum_A m_A \delta^{(3)}(\mathbf{x} - \mathbf{x}_A), \quad (88)$$

$$\nabla^2 V_i = -4\pi G \sum_A m_A v_A^i \delta^{(3)}(\mathbf{x} - \mathbf{x}_A), \quad (89)$$

one recovers the standard EIH Lagrangian with the coefficients shown in Eq. (31).

In the superfluid model, the scalar and optical sectors ensure that U behaves as in GR at 1PN order, while the dyon construction ensures that V_i has the dipolar structure $V_i \propto \epsilon_{ijk} J^j x^k / r^3$ (Cartesian components $\propto 1/r^2$) around individual spinning masses. The nontrivial work in this paper is to show that, once the vector interaction arising from overlapping translational wakes is projected onto an $1/r_{AB}$ kernel, the resulting coefficients of $\mathbf{v}_A \cdot \mathbf{v}_B$ and $(\mathbf{v}_A \cdot \mathbf{n}_{AB})(\mathbf{v}_B \cdot \mathbf{n}_{AB})$ match those of the EIH Lagrangian when the wake parameters are tuned to $a_H = 0$, $\alpha^2 = 3/4$, and $K = 2/\pi^2$; the scalar and optical sectors are fixed independently in Papers I and II.

D.3 PPN parameters and internal consistency

The PPN formalism provides a convenient way to check the internal consistency of the emergent metric across different sectors:

- Paper I fixed β_{eff} by demanding that the perihelion precession of nearly Keplerian orbits match GR. This depends primarily on the structure of g_{00} and its U^2 term.
- Paper II fixed γ_{eff} by matching light bending, Shapiro delay, and redshift, which depend on the combination of g_{00} and g_{ij} through the optical metric constructed from $N(r)$.
- This paper effectively fixes the vector-sector PPN parameter (the analogue of α_1 in some PPN conventions) by matching the EIH velocity-dependent tensor structure, which depends on g_{0i} and its relation to the flow.

In all three cases the matching conditions are applied to the *same* underlying medium response, whose acoustic (optical) and hydrodynamic (matter) projections can be packaged into consistent effective 1PN metric data once calibrated, parameterized by a handful of hydrodynamic response coefficients: the throat geometry (L/a), the scalar renormalization parameter (β in the defect Lagrangian), the polytropic index (n), and the longitudinal/transverse mixing in the dyon flow (α). The fact that a single choice of these parameters yields $\beta_{\text{eff}} = \gamma_{\text{eff}} = 1$ and reproduces the full EIH Lagrangian at 1PN order is the main consistency result of the series.

D.4 Sector-by-sector check

For completeness, we summarize the sector-by-sector mapping between metric components and EIH terms:

Scalar sector (g_{00}): The combination of the Newtonian potential, lag correction, and density-dependent mass $m(\Phi_{\text{loc}})$ fixes g_{00} and generates both the 1PN perihelion advance and the static G^2 three-body term in L_{EIH} , with $\beta_{\text{eff}} = 1$.

Optical sector (g_{ij}): The $n = 5$ refractive index profile $N(r)$ yields an effective spatial metric with $\gamma_{\text{eff}} = 1$ inferred from the optical metric/observables, reproducing the 1PN light-bending and Shapiro-delay coefficients and entering the $v^2 U$ terms in the EIH Lagrangian.

Vector sector (g_{0i}): The translational wake and its overlap kernel define the vector potential V_i and hence g_{0i} . Once the scalar ($q = 1$) [1] and optical ($n = 5$) [2] contributions are included, the EIH matching conditions fix the wake parameters to $a_H = 0$, $\alpha^2 = 3/4$, and $K = 2/\pi^2$, as summarized in Sec. 4.4. With this choice, the resulting velocity-dependent interaction matches the EIH tensor structure in Eq. (31), including the relative coefficients of $\mathbf{v}_A \cdot \mathbf{v}_B$ and $(\mathbf{v}_A \cdot \mathbf{n}_{AB})(\mathbf{v}_B \cdot \mathbf{n}_{AB})$.

Thus, starting from a hydrodynamic description of a single medium, one recovers the same metric data (U, V_i, β, γ) that underlie the standard EIH derivation in GR. The emergent metric of the superfluid toy universe is therefore EIH-equivalent to the GR metric at 1PN order, within the regimes and approximations considered in this series.

References

- [1] Norris, T. (2025). *Newtonian and 1PN Orbital Dynamics from a Superfluid Defect Toy Model* Zenodo. [doi:10.5281/zenodo.19449058](https://doi.org/10.5281/zenodo.19449058).
- [2] Norris, T. (2025). *Gravitational Optics and Soliton Geodesics in a Superfluid Defect Toy Model*. Zenodo. [doi:10.5281/zenodo.19449182](https://doi.org/10.5281/zenodo.19449182).